

Fangfang Xu

Master's Student in Computer Science and Technology

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Research Interests

Predictive and interventional computational biology, with a focus on modeling how cells and tissues respond to genetic and chemical perturbations. I am particularly interested in multimodal and spatial representations of biological states for perturbation-response prediction, virtual cell and tissue modeling, and target prioritization. Building on my experience in spatial omics and histopathology, I aim to develop computational models that connect observational biological measurements with experimentally testable predictions of intervention response.

Education

Xiamen University

Xiamen, China

M.S. in Computer Science and Technology (in progress)

Sep. 2024 - Jun. 2027 (expected)

GPA: 3.7/4.0. Specialization: Health Big Data and Intelligent Medicine.

Joint supervision: Chaoyong Yang, Liansheng Wang, and Jia Song.

Selected courses (/100): Matrix Theory (96), Machine Learning (94), Deep Learning (93), Computational Linguistics (91).

China University of Mining and Technology, Beijing

Beijing, China

B.Eng. in Remote Sensing Science and Technology

Sep. 2019 - Jun. 2023

GPA: 3.69/4.0.

Selected courses (/100): Advanced Mathematics (99), Probability and Mathematical Statistics (93), Linear Algebra (91).

Research Experience & Preprint

Spaceland: Histology-Guided Reconstruction of High-Resolution Whole-Organ 3D Molecular Atlases from Sparse Spatial Transcriptomics

Dec. 2024 – Present

Fangfang Xu*, Zhenfeng Zhuang*, Yun Zhu*, Bo Ying, Naiqiao Hou, Weiping Lin, Liansheng Wang, Chaoyong Yang, Jia Song. *bioRxiv preprint*, 2026. **Manuscript under revision at Nature Communications.**

Preprint: [10.64898/2026.07.23.739686](https://doi.org/10.64898/2026.07.23.739686) Code: github.com/xufangfang408/Spaceland

First author. Conceived the research project, designed the model, conducted the majority of benchmarking experiments, and wrote the manuscript.

Method: Developed a histology-conditioned continuous molecular field model that combines foundation-model H&E representations with inter-section deformation modeling to infer gene expression across sparsely measured tissue volumes.

Results: Reconstructed continuous 3D gene-expression fields on an 8- μ m grid from spot-level spatial transcriptomics with molecular sampling gaps up to 400 μ m; evaluated across mouse olfactory bulb, mouse hemibrain, and planarian regeneration datasets.

Spaceland-omics | Histology-guided spatial multi-omics modeling

Jun. 2026 - Present

Extending Spaceland from transcriptomics to multimodal molecular prediction by using serial histology as a shared spatial context. The project investigates whether a common tissue representation can support continuous modeling of transcriptomic and proteomic states across sparsely measured sections.

OmicsBind | Histology-anchored multimodal representation learning

Jul. 2026 - Present

Developing an ImageBind-inspired framework that aligns spatial molecular modalities through a shared histology anchor. Each molecular modality is paired only with H&E rather than directly with every other modality, enabling multimodal representation learning from partially paired spatial-omics datasets.

Selected Honors

Second Prize, National Final

May 2023

14th Chinese Mathematics Competitions for College Students, Non-Mathematics Category.

First Prize, Preliminary Round

Dec. 2022

14th Chinese Mathematics Competitions for College Students, Non-Mathematics Category; ranked 36th in Beijing.